

⇒ RK Method:-

its accuracy is better than others
it is 10 times better than Euler:-

$$\begin{aligned} \Rightarrow k_1 &= hf(n_n, y_n) \\ k_2 &= hf\left(n_n + \frac{h}{2}, y_n + 2hk_1\right) \end{aligned}$$

$$k_3 = hf\left(n_n + \frac{h}{2}, y_n + 2k_2\right)$$

$$k_4 = hf(n_{n+1}, y_n + k_3)$$

$$y_{n+1} = y_n + \frac{h}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

⇒ Practice problem:-

ex1, $\frac{dy}{dx} = x+y$ where $h=0.2$
 $y(0)=1$

$y = ??$
 $y_1 = y_0 + k_1$

$$k_1 = \frac{1}{6} [k_1 + 2k_2 + 3k_3 + k_4]$$

~~$k_1 = hf(n_0, y_0)$~~

$$k_1 = hf(n, y)$$

$$k_1 = 0.2(0+1) = 0.2(1) = 0.2 \rightarrow [k_1]$$

$$k_2 = hf\left(n_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_2 = 0.2\left(0 + \frac{0.2}{2} + 1 + \frac{0.2}{2}\right)$$

$$[k_2 = 0.24]$$

$$k_3 = hf \left(n_0 + \frac{h}{2}, y_0 + \frac{k_2}{2} \right)$$

$$k_3 = 0.2 \left(0 + \frac{0.2}{2} + 1 + \frac{0.24}{2} \right)$$

$$k_3 = 0.2 [0.1 + 1.12]$$

$$\boxed{k_3 = 0.244}$$

$$k_4 = hf \left(n_0 + h, y_0 + k_3 \right)$$

$$k_4 = 0.2 [0 + 0.2 + 1 + 0.244]$$

$$\boxed{k_4 = 0.288}$$

now

$$k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$k = \frac{1}{6} [0.2 + 2(0.24) + 2(0.244) + 0.288]$$

$$k = \frac{1}{6} [1.456]$$

$$= 0.242$$

$$y_1 = [y_0 + k]$$

$$= \boxed{1.242}$$

Ans

$$\text{ex} \Rightarrow \frac{dy}{dn} = y - n, \quad y(0) = 1 \quad (h = 0.1)$$

$$y = ??$$

$$y_1 = y_0 + k_1$$

$$k_1 = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$k_1 = hf[n_0, y_0]$$

$$k_1 = h[-n_0 + y_0]$$

$$k_1 = h[-0 + 1]$$

$$k_1 = 0.1(1)$$

$$k_1 = 0.1$$

Now for second iteration.

$$k_2 = hf\left[n_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right]$$

$$k_2 = 0.1 \left[-0 + \frac{0.1}{2} + 1 + \frac{0.1}{2}\right]$$

$$k_2 = 0.11$$

$$k_3 = hf\left[n_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right]$$

$$k_3 = 0.1 \left[-0 + \frac{0.1}{2} + 1 + \frac{0.11}{2}\right]$$

$$k_3 = 0.1 \left[\right]$$

$$= 0.1105$$

$$k_4 = hf[n_0 + h, y_0 + k_3]$$

$$k_4 = hf[0 + 0.1 + 1 + 0.1]$$

$$k_4 = 0.1[1.2]$$

$$k_4 = 0.12$$

$$y_1 = y_0 + k_1 = 1 + 0.1 = 1.1$$

$$k_1 = 0.1$$

$$y_1 = 1.1$$

⇒ Solve the system of D.E using Laplace transform

$$\rightarrow \frac{dx}{dt} - y = e^t \rightarrow (i)$$

$$y(0) = 0$$
$$x(0) = 1.$$

$$\frac{dy}{dt} + x = \sin t \rightarrow (ii)$$

sol

$$x' - y = e^t$$

$$s x(s) - x(0) - y(s) = \frac{1}{s-1}$$

$$s x(s) - 1 - y(s) = \frac{1}{s-1}$$

$$s x(s) - y(s) = \frac{1}{s-1} + 1$$

$$s x(s) - y(s) = \frac{1+s-1}{s-1}$$

$$s x(s) - y(s) = \frac{s}{s-1} \rightarrow \text{eq (i)}$$

$$y' + x = \sin t$$

$$s y(s) - y(0) + x(s) = \frac{1}{s^2+1}$$

$$s y(s) - 0 + x(s) = \frac{1}{s^2+1}$$

$$s y(s) + x(s) = \frac{1}{s^2+1} \Rightarrow \text{eq (ii)}$$

$$s^n(s) - y(s) = \frac{s}{s-1} \rightarrow \text{eq (i)}$$

$$s^n(s) + s y(s) = \frac{1}{s^2+1} \Rightarrow \text{eq (ii)}$$

$$\cancel{s^n(s)} - y(s) = \frac{s}{s-1}$$

$$-\cancel{s^n(s)} + \cancel{s^2 y(s)} = -\frac{s}{s^2+1}$$

$$-s^2 y(s) - y(s) = -\frac{s}{s^2+1} + \frac{s}{s-1}$$

$$y(s) \cdot [s^2+1] = -\frac{s}{s^2+1} + \frac{s}{s-1}$$

$$y(s) = \frac{s}{(s^2+1)^2} - \frac{s}{(s-1)(s^2+1)^2}$$

Now

$$\frac{s(s-1) - s(s^2+1)}{(s^2+1)^2 (s-1)}$$

$$\frac{s^2 - s - s^3 - s}{(s^2+1)^2 (s-1)}$$

$$\frac{s^2 - 2s - s^3}{(s^2+1)^2 (s-1)}$$

$$y(s) \frac{-s^3 + s^2 - 2s}{(s^2+1)^2 (s-1)} = \frac{A}{(s-1)} + \frac{Bn+C}{(s^2+1)} + \frac{Dn+E}{(s^2+1)^2}$$

$$-s^3 + s^2 - 2s = A[s^2+1] + Bs + c[s-1][s^2+1] + Ds + E$$

$$= A[s^4 + 2s^2 + 1] + Bs + c[s^3 + s - s^2 - 1] + Ds + E[s-1]$$

$$= As^4 + 2As^2 + A + Bs + c[s^3 - s^2 - 1] + Ds + E[s-1]$$

$$= As^4 + 2As^2 + A + Bs^3 + Bs + cs^3 - cs^2 - c + Ds^2 - Ds + Es - E$$

$$As^4 = 0$$

$$Bs^3 = -s^3$$

$$B = -1 \quad | \quad A = 0$$

$$2As^2 + cs^2 + Ds^2 = s^2$$

$$\rightarrow As^4 + 2As^2 + A + Bs^3 + Bs^2 - Bs + c + cs - cs^2 - c + Ds^2 - Ds + Es - E$$

$$(A+B=0)$$

$$(-B+C = -1)$$

$$2A+B+C+D=1$$

$$-2 = -B+C-D+E$$

$$A - C - E = 0$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ 2 & 1 & 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 & -2 \\ 1 & -1 & -1 & 0 & -1 & 0 \end{bmatrix}$$

Now Apply,

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \\ 2 & 1 & 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 & -2 \\ 1 & 0 & -1 & 0 & -1 & 0 \end{array} \right] \begin{array}{l} -R_2 \\ R_3 - 2R_1 \\ 1 - 2(1) = -1 \\ 1 - 2(0) = 1 \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 & -2 \\ 1 & 0 & -1 & 0 & -1 & 0 \end{array} \right] R_3 + R_2$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & -1 & 1 & -1 & 1 & -2 \\ 1 & 0 & -1 & 0 & -1 & 0 \end{array} \right] R_5 \leftrightarrow R_1$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 1 & 0 & -1 & 0 & -1 & 0 \end{array} \right] R_3 + R_2 \quad \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 1 & 0 & -1 & 0 & -1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{array} \right] \quad R_3 - R_2$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{array} \right] \quad R_3 + R_2$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & -2 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{array} \right] \quad -\frac{1}{2}R_3$$

$$\frac{0}{(s+1)} + \frac{0-1}{(s+1)} + \frac{2s+1}{(s+1)^2}$$

$$\left[\begin{array}{ccccc|c} \textcircled{1} & 1 & 0 & 0 & 0 & 0 \\ 0 & \textcircled{1} & -1 & 0 & 0 & 1 \\ 0 & 0 & \textcircled{1} & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{array} \right] \quad \left\{ \begin{array}{l} \frac{-1}{s+1} + \frac{2s+1}{(s+1)^2} \end{array} \right.$$

Now Replace:

$$R_5 + R_4 \quad \boxed{A=0}$$

$$A+B=0$$

$$B-C=1$$

$$B+1=1 \quad \boxed{B=0}$$

$$C+\frac{1}{2}E=-\frac{1}{2}$$

$$\boxed{D=2}$$

$$C+\frac{1}{2}E=-\frac{1}{2}$$

$$\boxed{E=1}$$

$$C=-\frac{1}{2}-\frac{1}{2}E$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right]$$